Comparative Benchmark Report — Smart AI Router

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Fixed Dataset: LexGLUE Benchmark (Fixed)
Fixed Downstream LLM: Gemini Flash (Fixed)
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1. Executive Summary

This empirical report presents a comparative evaluation of context routing principles desi

2. Comparative Performance Matrix

Routing Principle	Precision	Recall	F1 Score	NDCG@3	Compression %	Token S			
bm25	100.0%	100.0%	100.0%	0.9000	50.0%	50.0%	80.0%	0.0010	$0.00
tfidf	100.0%	100.0%	100.0%	0.9000	50.0%	50.0%	80.0%	0.0010	$0.0
embedding	100.0%	100.0%	100.0%	0.9000	50.0%	50.0%	80.0%	0.0010	

3. Visual Performance Comparisons

![Retrieval Quality Comparison](/home/imchaul/Coding Files/Python/maddy_project/data/repor
![Compression vs Accuracy](/home/imchaul/Coding Files/Python/maddy_project/data/reports/fi

4. Key Findings & Discussion

- Optimal Router: The bm25 routing principle achieved the highest overall F1 score (100.0%
- Cost-Accuracy Tradeoff: Higher context compression significantly reduces downstream prom
- Latency Impact: Neural and vector similarity routing principles introduce minor initial

5. Conclusion

Context routing principles provide measurable token savings and latency reductions for LLM